

2016.0 RANGE ROVER (LG), 303-01

ENGINE - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

CRANKSHAFT FRONT SEAL - TDV6 3.0L DIESEL (G1509139)

REMOVAL AND INSTALLATION

12.21.14	OIL SEAL - CRANKSHAFT - FRONT - RENEW	3000 CC, TDV6	3.6	USED WITHINS	+
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SPECIAL TOOL(S)



303-1120

Remover,
Crankshaft Front
Seal



E52717

303-1121

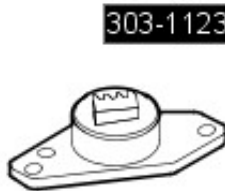
Installer,
Crankshaft Seal



E54545

303-1122

Installer, Crankshaft
Front Seal



E 54546

303-1123

Locking Tool,
Flywheel



E64849

303-D121

Puller, General
Purpose

PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME
Installation Step 1	Renew Part	Crankshaft front oil seal
Installation Step 6	Renew Part	Crankshaft pulley bolt
Installation Step 8	Renew Part	Crankshaft pulley friction washer
Installation Step 8	Renew Part	Crankshaft pulley bolt

REMOVAL

NOTE:

Some variation in the illustrations may occur, but the essential information is always correct.

1. Disconnect the battery ground cable.
Refer to: [Battery](#) (414-01 Battery, Mounting and Cables, Removal and Installation).

2.

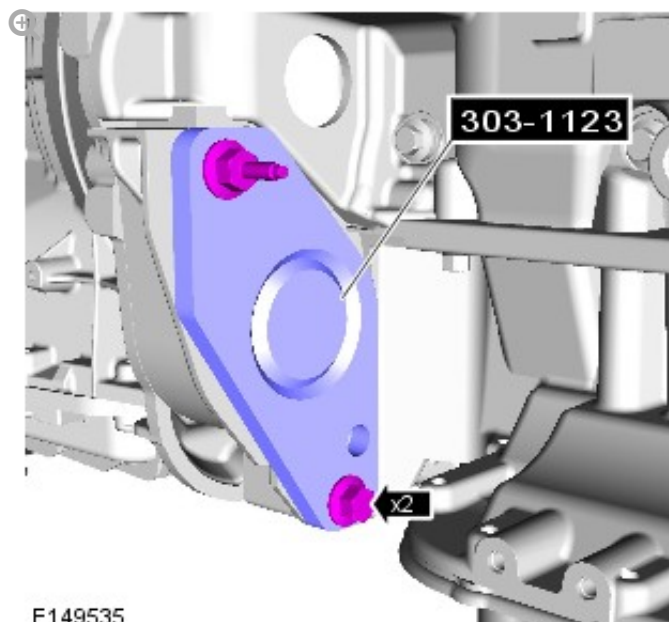
WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

3. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).

4.

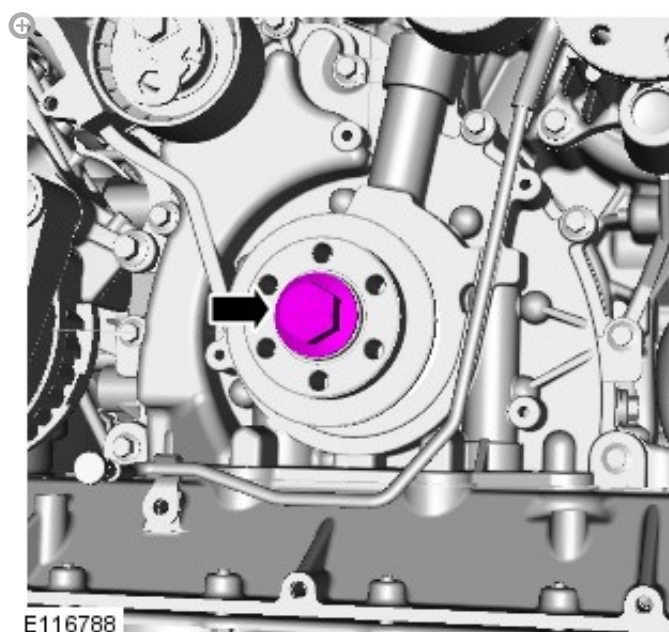


E149535

Install the special tool.

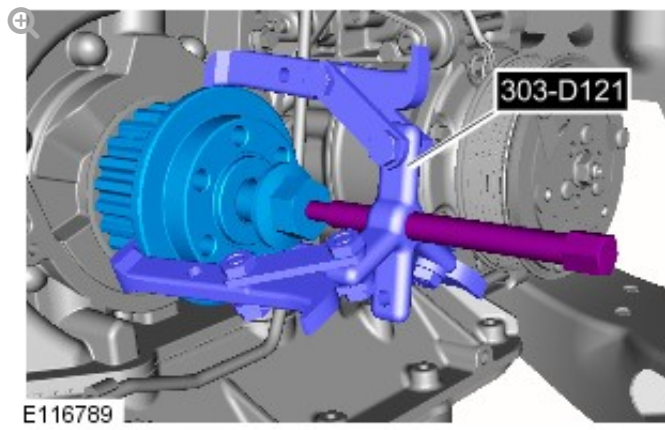
Special Tool(s): [303-1123](#)

5.



E116788

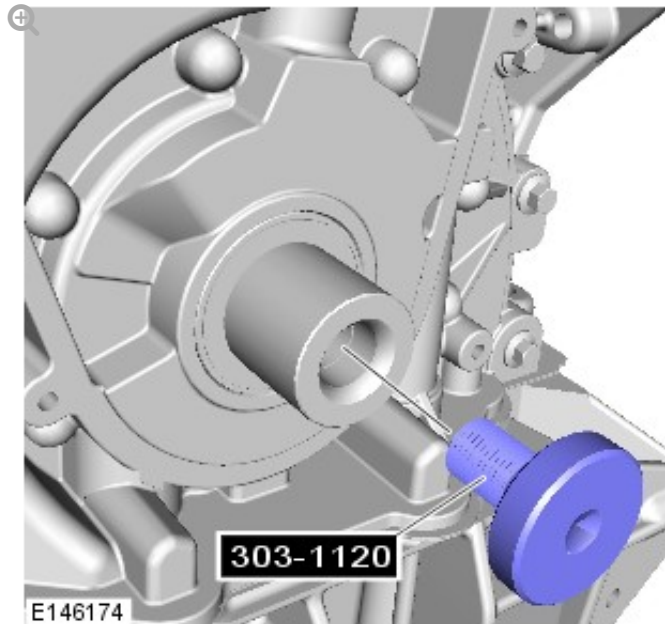
6.



Install the special tool.

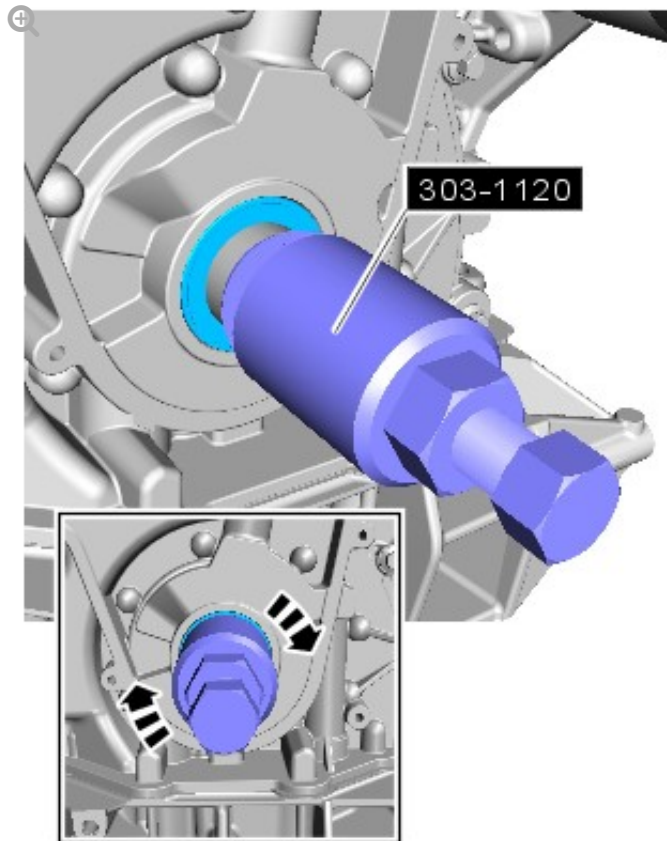
Special Tool(s): [303-D121](#)

7.



Install the special tool to the crankshaft.

8.

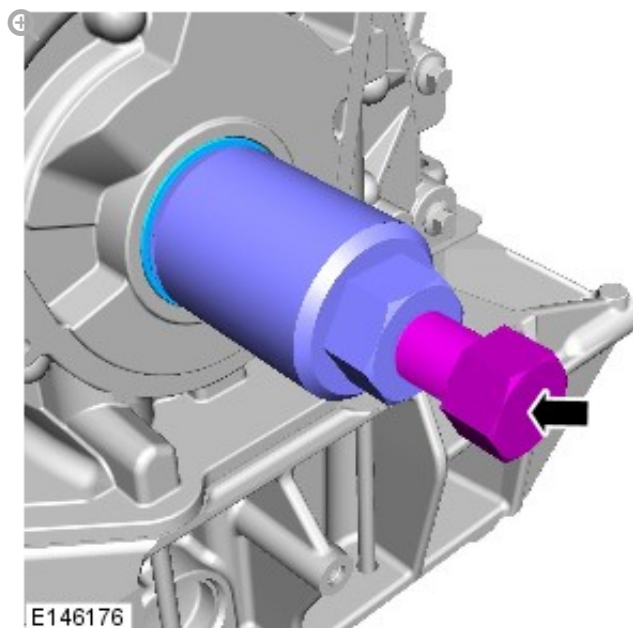


E146175

Install the special tool to the crankshaft, rotate the special tool to engage the crankshaft front oil seal as shown.

Special Tool(s): [303-1120](#)

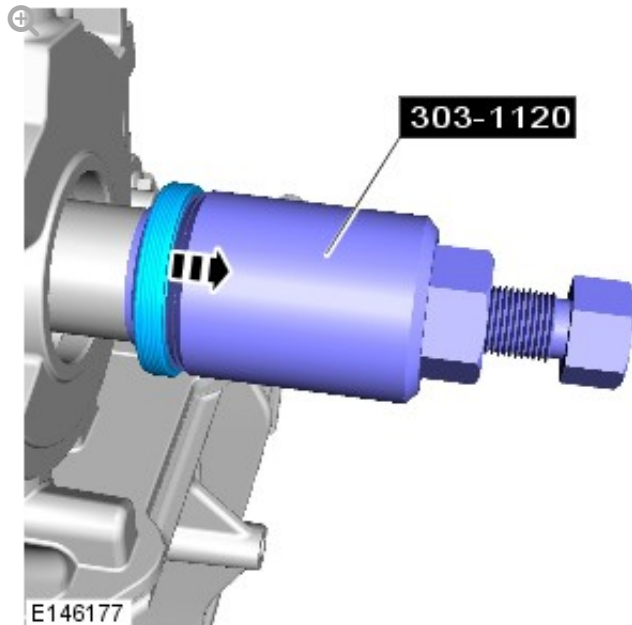
9.



E146176

Rotate the special tool bolt.

10.



Using the special tool, remove and discard the crankshaft front oil seal.

INSTALLATION

1.

CAUTION:

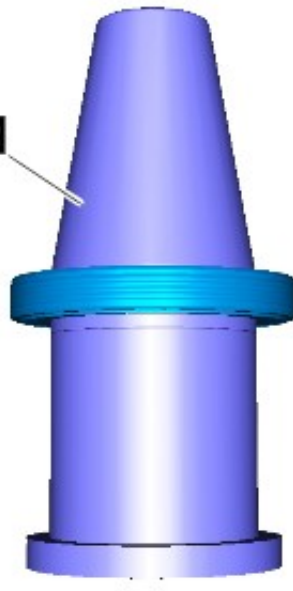
Do not use any lubricant on the crankshaft front seal, special tools or the crankshaft. Failure to follow this instruction may result in damage to the vehicle.

NOTE:

Make sure that all the component mating faces are clean.



303-1122



E116752

Install a new crankshaft front seal to the special tool.

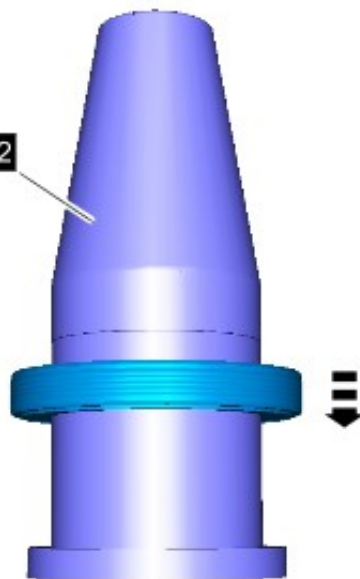
Special Tool(s): 303-1122

Renew Part: [Crankshaft front oil seal](#).

2.



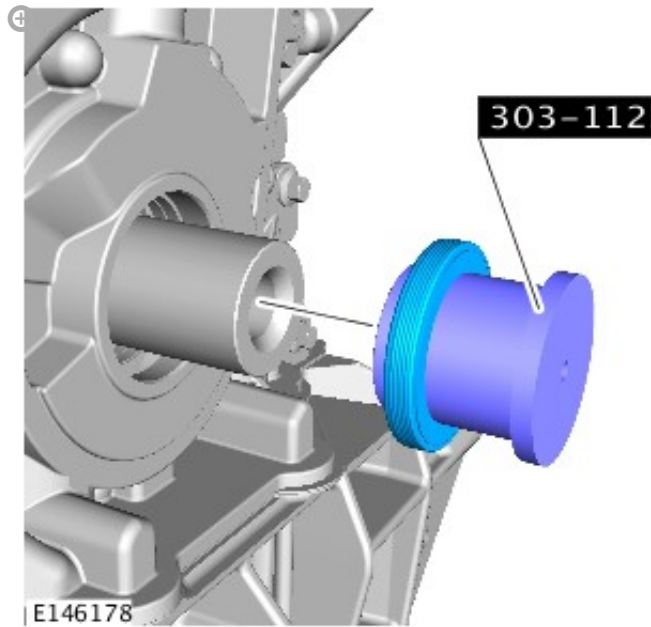
303-1122



E116753

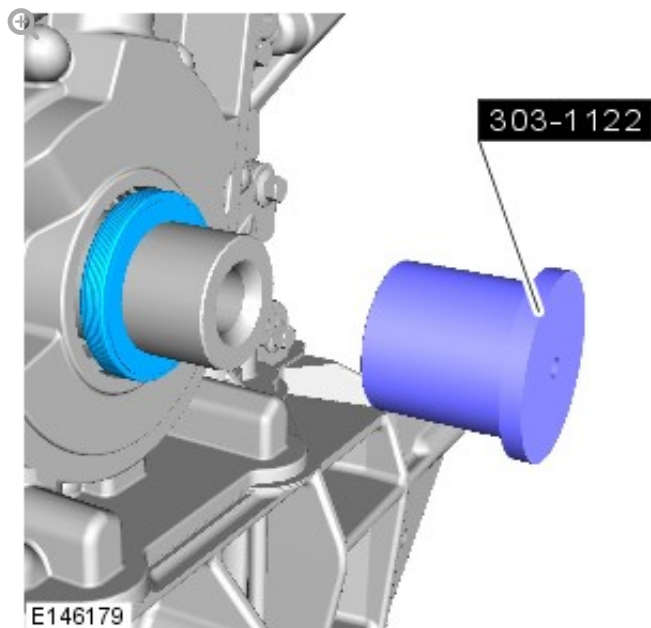
Reposition the crankshaft front seal along the special tool.

3.



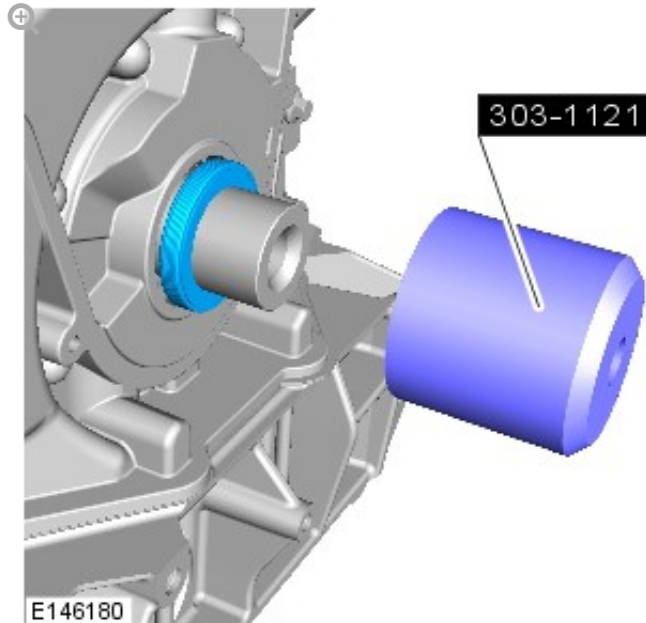
Install the special tool to the crankshaft.

4.



Install the special tool to the crankshaft.

5.



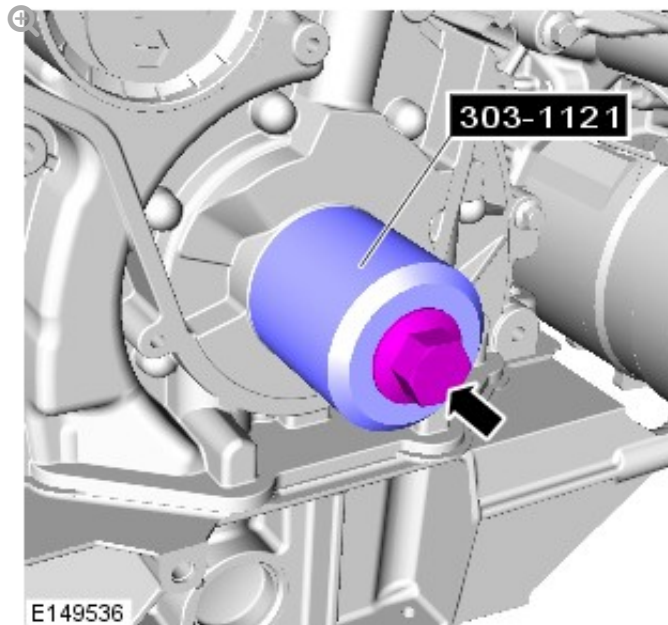
Install the special tool to the crankshaft.

Special Tool(s): 303-1121

6.

CAUTIONS:

- Make sure the seal is installed correctly.
- Make sure that the seal is seated 1mm under flush.
- Discard the bolt.



Using the special tool, install the crankshaft front seal.

Renew Part: [Crankshaft pulley bolt](#).

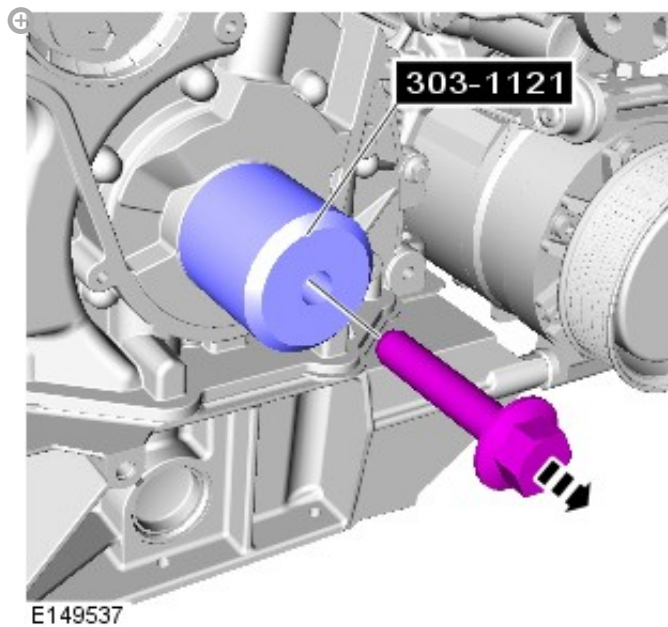
7.

CAUTION:

Make sure the seal is installed correctly.

NOTE:

Wait 10 seconds before removing the crankshaft seal installation tool.



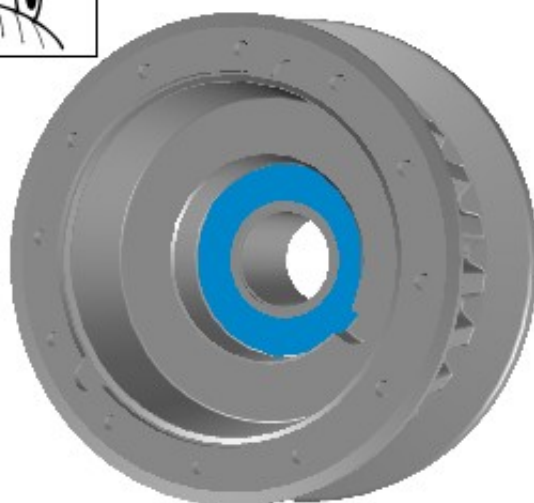
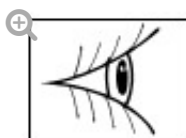
Remove the special tool.

Special Tool(s): [303-1121](#)

8.

CAUTION:

Make sure that the pulley washer is correctly seated before installing the pulley.



E178218

Renew Part: [Crankshaft pulley friction washer.](#)

Renew Part: *Crankshaft pulley bolt.*

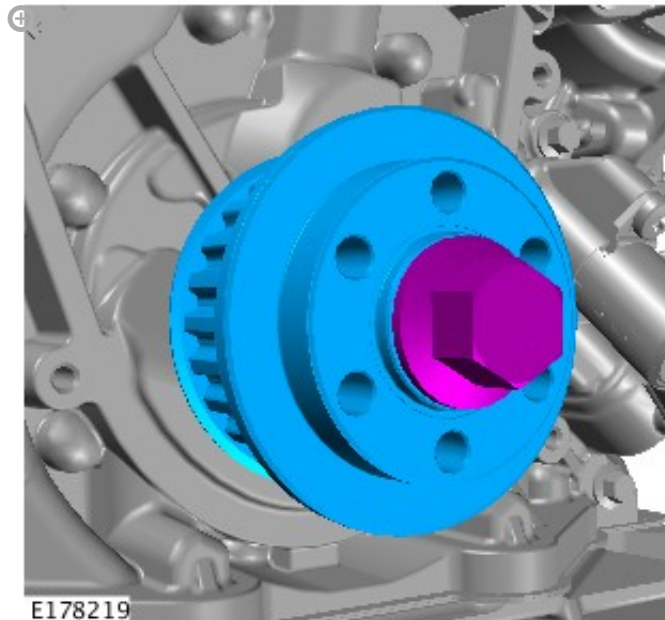
9.

CAUTIONS:

- Do not lubricate the components.
- Make sure that a new bolt is installed.

NOTE:

Do not tighten at this stage.

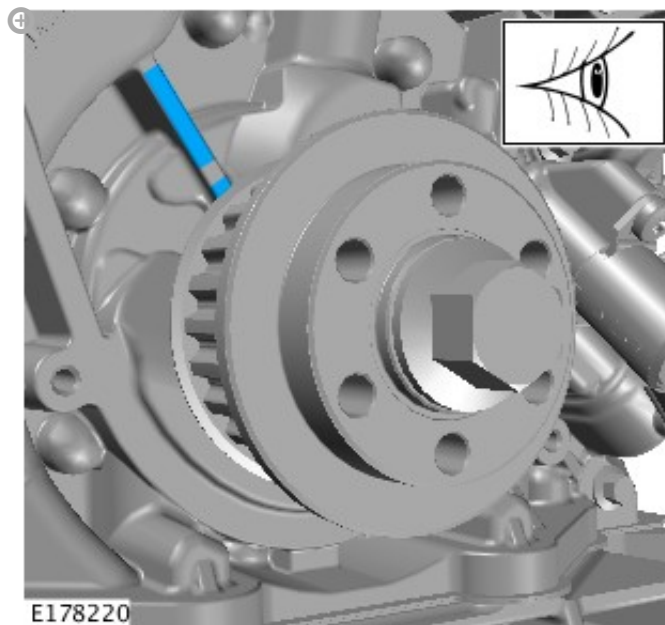


Install crankshaft pulley.

10.

CAUTION:

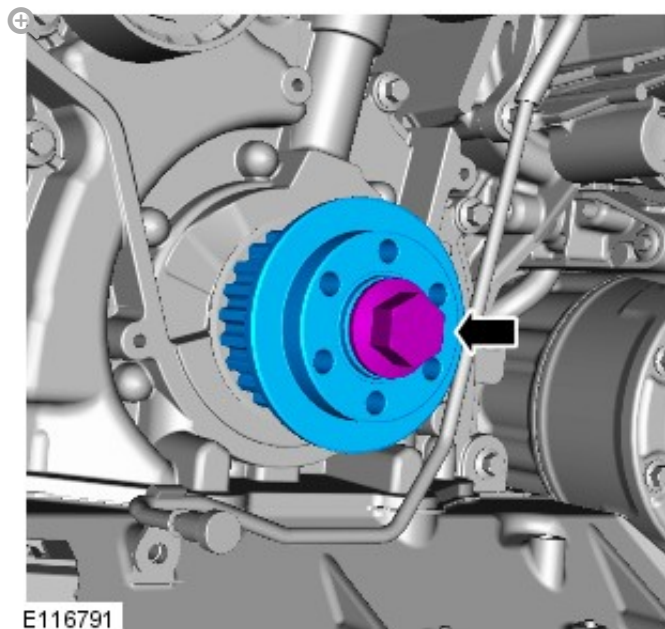
Make sure the alignment marks are aligned.



11.

CAUTION:

Do not lubricate the components.



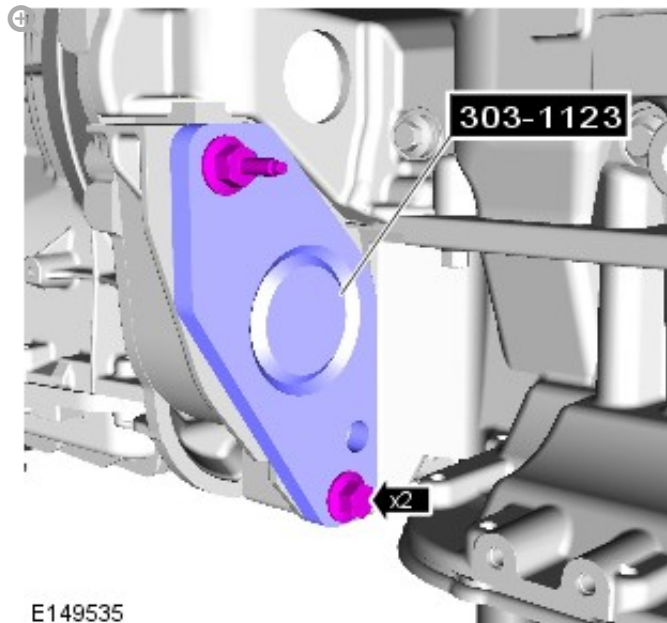
Torque:

M16 retaining bolt

Stage 1: 100 Nm

Stage 2: 35°

12.



Remove the special tool.

13. Refer to: [Timing Belt](#) (303-01F Engine - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Removal and Installation).
14. Connect the battery ground cable.
Refer to: [Battery](#) (414-01 Battery, Mounting and Cables, Removal and Installation).